INCH POUND
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22 February 1991
SUPERSEDING
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10 May 1974

MILITARY SPECIFICATION

TUBING, HEAT SHRINKABLE

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for heat-shrinkable tubing intended for use in rocket motor igniters.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5970
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-585	-	Boxes, Wood, Wirebound
PPP-B-591	-	Boxes, Fiber Board, Wood-Cleated
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-T-76	-	Tape, Packaging, Paper (for Carton Sealing)

MILITARY

MIL-P-116	-	Preservation, Methods of
MIL-I-7444	-	Insulation Sleeving, Electrical, Flexible
MIL-L-10547	-	Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible

STANDARDS

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-1190	-	Minimum Guidelines for Level C Preservation, Packing and Marking

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents from a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN SOCIETY FOR TESTING MATERIALS (ASTM)

ASTM D 570	-	Water Absorption of Plastics, Standard Test Method for
ASTM D 876	-	Nonrigid Vinyl Chloride Polymer Tubing, Standard Method of Testing

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Tubing. The material specified herein shall be supplied in the form of tubing with an inside diameter (ID) measuring not less than 0.900 inch.

3.2 Physical properties. The physical properties of the material shall be as specified herein.

3.2.1 Dimension recovery. The dimension of the tubing in the expanded form, as supplied, and the dimensions after heating to 130 ± 10° Celsius (C) shall be not greater than 0.450 inch ID and 0.020 ± 0.003 inch wall thickness (see 4.7.1).

3.2.2 Longitudinal shrinkage. In reaching its contracted dimensions, the tubing shall not shrink greater than 15 percent longitudinally (see 4.7.2).

3.2.3 Tensile strength. The tensile strength of the tubing shall be not less than 2500 pounds force per square inch (lbf/in^2) (see 4.7.3).

3.2.4 Ultimate elongation. The ultimate elongation shall be not less than 200 percent (see 4.7.4).

3.2.5 Brittleness temperature. The tubing shall not crack at -55° ± 3°C (see 4.7.5).

3.2.6 Heat aging. The tubing shall not crack or split after being continuously subjected to a temperature of 130° ± 3°C for 168 hours (seven days) (see 4.7.6).

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3.2.7 Corrosion. The tubing shall not accelerate corrosion of any metals with which it may come in contact (see 4.7.7).

3.2.8 Water absorption. Water absorption shall not vary the original weight by greater than ± 0.5 percent (see 4.7.8)

3.2.9 Heat shock. The tubing shall not drip, flow, or crack when continuously subjected to a temperature of 130° ± 5°C for a period of four hours (see 4.7.9).

3.2.10 Fungus resistance. The tubing shall not support fungus growth (see 4.7.10).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as quality conformance inspection (see 4.3).

4.3 Quality conformance inspection. Quality conformance inspections shall be as specified in table I.

4.4 Lot size. A lot shall be as specified in the contract or order with the following additional requirements: if manufacture is by batch process, each batch, blend of batches, or fraction thereof, shall constitute a lot (see 6.2).

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TABLE I. Quality conformance inspection.

Inspection	Requirement paragraph	Test paragraph
Dimensional recovery	3.3.1	4.7.1
Longitudinal shrinkage	3.3.2	4.7.2
Tensile strength	3.3.3	4.7.3
Ultimate elongation	3.3.4	4.7.4
Brittleness temperature	3.3.5	4.7.5
Heat aging	3.3.6	4.7.6
Corrosion	3.3.7	4.7.7
Water absorption	3.3.8	4.7.8
Heat shock	3.3.9	4.7.9
Fungus resistance	3.3.10	4.7.10

4.5 Sampling. Unless specified in the contract or order, sampling for inspection or conformance shall be as specified in table II.

4.6 Sampling units. One length of tubing shall be considerd a unit for sample size considerations and related testing purposes.

TABLE II. Sampling for quality conformance.

Lot Size	Sample Size
2 - 15	2
16 - 25	3
26 - 90	5
91 - 150	8
151 - 280	13
281 - 500	20
501 - 1200	32
1201 - 3200	50
3201 - 10,000	80
10,001 - 35,000	125
35,001 - 150,000	200
In all cases: Acceptance number is ZERO Rejection number is ONE	

4.7 Test methods.

4.7.1 Dimensional recovery. The ID of three six-inch specimens of tubing shall be measured using plug gages and a micrometer. These samples shall be placed for one minute in a 130° ± 10°C oven. Upon removal, they shall be cooled to room

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temperature. The ID shall then be remeasured, using plug gages and micrometers. The test shall be run concurrently with 4.7.2, using the same samples to determine compliance with 3.2.1.

4.7.2 Longitudinal shrinkage. The length of the three specimens used in 4.7.1 shall be carefully measured to ± 0.0625 inch prior to being placed in the oven as specified in 4.7.1. Upon removal from the oven and cooling to room temperature, the length shall be remeasured and the shrinkage calculated as a percentage of the original length in accordance with 3.2.2.

4.7.3 Tensile strength. Tensile strength shall be determined in accordance with ASTM D 876 and 3.2.3.

4.7.4 Ultimate elongation. Ultimate elongation of the tubing shall be determined in accordance with ASTM D 876 and 3.2.4.

4.7.5 Brittleness temperature. Three specimens, each approximately 18 inches long, shall be conditioned at -55° ± 3°C for four hours. A fixed mandrel measuring 0.37 ± 0.03 inch in diameter shall be conditioned at this same temperature. Upon completion of this conditioning, and at this same temperature, the specimens shall be rapidly wrapped about the mandrel for not less than three complete wraps. The speed of wrapping shall be approximately two seconds per 360° wrap and shall be in accordance with 3.2.5.

4.7.6 Heat aging. Three specimens, each approximately six inches in length, shall be tested at a temperature of 130° ± 3°C. The specimens shall be placed in a uniformly-heated oven, of the forced draft or convection type having sufficient capacity to produce and maintain the specified temperatures. The specimens shall be removed at the end of 168 hours (seven days) and cooled to room temperature. The specimens shall then be bent through 180° over a 0.37 ± 0.03 inch mandrel. Any side cracking occurring because of the flattening of the specimen on the rod shall be disregarded. Heat aging will be conducted to determine compliance with 3.2.6.

4.7.7 Corrosion. Corrosion shall be tested in accordance with MIL-I-7444 and 3.2.7.

4.7.8 Water absorption. Water absorption test shall be in accordance with ASTM D 570 and 3.2.8.

4.7.9 Heat shock. Three specimens, each approximately four inches long, shall be conditioned for four hours at 130° ± 5°C. While at that temperature and upon removal, they shall be examined for evidence of flow, dripping, or cracking to determine compliance with 3.2.9.

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4.7.10 Fungus resistance. Fungus resistance shall be determined in accordance with MIL-I-7444 and 3.2.10.

4.7.11 Examination. Packaging and marking of material containers shall be inspected for conformance to the requirements of Section 5.

4.8 Inspection of packaging. Except when commercial packaging is specified, the sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification shown in Section 5. The inspection of marking for shipment and storage shall be in accordance with MIL-STD-129. The inspection of commercial packaging shall be as specified in the contract (see 6.2).

5. PACKAGING

5.1 Packaging requirements. Packaging shall be Level A or C as specified (see 6.2).

5.1.1 Level A. The tubing shall be packaged in accordance with Method III of MIL-P-116.

5.1.2 Level C. The tubing shall be packaged in accordance with MIL-STD-1190.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The tubing, packaged in accordance with 5.1.1, shall be packed in overseas-type wood-cleated fiberboard, nailed wood, wirebound wood, corrugated or solid fiberboard, wood-cleated paper overlaid or wood-cleated plywood boxes conforming to PPP-B-591; PPP-B-621; PPP-B-585; PPP-B-636, CF or SF; or PPP-B-601, respectively at the option of the contractor. Unless otherwise specified, shipping containers shall have caseliners conforming to MIL-L-10547 and shall be closed and sealed in accordance with the appendix thereto. Caseliners for boxes conforming to PPP-B-636 may be omitted provided all joints and corners of the boxes are sealed with not less than 1.5 inch wide tape conforming to PPP-T-76. Box closures shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood boxes shall be not greater than 200 pounds. Fiberboard boxes shall be not greater than the gross weight limitation and shall conform to the special requirements of the applicable box specification.

5.2.2 Level B. Same as Level A.

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5.2.3 Level C. The tubing, packaged in accordance with 5.1.2, shall be packed in containers in a manner to insure safe delivery and acceptance at destination.

5.3 Marking. Containers shall be marked in accordance with MIL-STD-129. Marking shall include, but not be limited to, the following information:

a. Manufacturer's name.

b. Material trade name and size.

c. Lot number.

d. Date of manufacture.

e. Number and revision letter of this specification.

f. Storage requirements (if any).

g. Toxicity precautions, if required.

h. Net weight.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This material is intended for use as a physical barrier for retaining pyrotechnic pellets in the basket of a rocket motor igniter.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

c. Lot size requirements (see 4.4).

d. Requirements for preservation, packing and marking (see 5.1, 5.2 and 5.3).

e. Conditions for quality conformance inspection (see 4.3).

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f. Request for copies of laboratory analysis.

g. Request for storage conditions and toxicity precautions.

6.3 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this document.

6.4 Subject term (keyword) listing.

Barrier, physical
Conduit, reducible
Cylinder, collapsible
Sleeve, insulating

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - MI

Review Activity:
DLA-GS

Preparing Activity:
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